AI HACKATHON · CASE BRIEF

# Automating the Learning Support Tutor's Workday

A current-state process brief: the daily duties of a Learning Support Tutor at a mathematics tutoring centre, and the opportunity to automate them with AI.

**PARTICIPANT BRIEFING DOCUMENT**

Role in focus: Learning Support Tutor · Weekday shift, 14:00–19:00

Prepared by AI Prototyping Studio, University of Melbourne

## 01 CASE OVERVIEW

This brief describes the daily duties of a Learning Support Tutor at a mathematics tutoring centre. Every step in Section 4 is a real activity the tutor performs on a normal weekday afternoon. Together they form the current-state workflow that teams will analyse.

| CASE AT A GLANCE | |
|---|---|
| **Role in focus** | Learning Support Tutor |
| **Organisation** | A mathematics tutoring centre |
| **Shift** | Weekdays, 14:00–19:00 |
| **Core tools** | Work WeChat, Google Classroom, Google Drive, and spreadsheets for progress and exam records |
| **Records maintained** | Marking Sheet **Exhibit 1**, Student Progress Sheet **Exhibit 2**, Student Exam Statistics **Exhibit 3** |
| **Workflow** | 5 phases and 9 numbered steps, with three tutoring scenarios (Section 4) |

## 02 THE CHALLENGE

YOUR TASK

**Identify the tasks in this workflow that can be partially automated with AI, and prototype a solution.**

## 03 KEY TERMS

| TERM | MEANING IN THIS BRIEF |
|---|---|
| **Module** | A unit of the centre's course. Students study it through recorded lessons, learning materials and worked examples, and finish with a Module Test. |
| **Demo** | A designated set of example questions that a student completes independently (without recordings, answers or notes) to show they have mastered a Module before sitting the Module Test. |
| **Module Test** | The centre's assessment for a Module, marked by the tutor. A student who does not pass may be scheduled for a Retest. |
| **SAC** | School-Assessed Coursework: internal assessments set by the student's own school under the Victorian Certificate of Education (VCE). |
| **Practice SAC** | A mock SAC, provided by the school or by the centre, used to rehearse the format and requirements of the real assessment. |
| **VCAA mock exam** | A practice exam based on a past paper from the Victorian Curriculum and Assessment Authority (VCAA). |
| **Google Classroom / Google Drive** | Platforms used by the centre to distribute learning materials and store student work and records. |
| **Student Service WeChat** | The centre's work WeChat account used to communicate with students and parents. |

## 04 CURRENT-STATE WORKFLOW

The tutor's shift runs in five phases. The overview below summarises each phase; the detailed steps follow.

| PHASE | TIME | FOCUS | KEY ACTIVITIES | STEPS |
|---|---|---|---|---|
| 1 | 14:00–14:15 | **Start of shift and preparation** | Open systems and records | 1 |
| 2 | 14:15–14:30 | **Reviewing the day's situation** | Timetable, attendance changes, student progress review | 2–5 |
| 3 | 14:30–16:00 | **Routine duties (non-tutoring)** | Student and parent messages; marking and record updates **Exhibits 1–3** | 6–8 |
| 4 | 16:00–18:00 | **Routine duties (tutoring and Q&A)** | Module study, SAC preparation, exams | 9 |
| 5 | 18:20–18:30 | **Daily work report** | Manager sign-off and summary of the day | — |

### PHASE 1 14:00–14:15 Start of shift and preparation

1. Open the systems and equipment needed for the day, including the work WeChat account, Google Classroom, Google Drive, student learning-progress records **Exhibit 2**, exam records **Exhibit 3** and any other relevant worksheets.

### PHASE 2 14:15–14:30 Reviewing the day's situation

2. Check the day's timetable and the on-site student status whiteboard. Confirm which students are attending, their assigned teachers, lesson times, classrooms, and whether each lesson is online or in person.
3. Confirm whether any student has requested leave, switched to an online lesson, rescheduled at short notice or has other special arrangements, and make sure this information has been passed on to the relevant teachers.
4. For students expected on site, review their recent learning progress in advance **Exhibit 2**: current position in the course, incomplete Tests, upcoming SACs, recent mistakes and any unresolved questions.
5. If students are already on site, have a brief conversation with them about their recent schoolwork, exams, SACs or homework. Adjust the day's study plan as needed and collect the students' needs and questions.

### PHASE 3 14:30–16:00 Routine duties (non-tutoring)

#### A. Handling student WeChat messages

6. Reply to messages from students and parents, including student questions, bookings for Q&A sessions, homework and Test status, parent feedback and other study-related matters.
7. Respond promptly to questions that can be resolved directly. Questions that involve in-depth academic content, are large in scope or fall outside the tutor's authority should be organised and either scheduled for a Q&A session or handed over to the relevant teacher or department, together with an explanation of the situation.

### B. Marking and updating student learning records

8. Mark the Tests previously submitted by students. After marking, record the results in the Marking Sheet **Exhibit 1**, the Student Progress Sheet **Exhibit 2** and the Student Exam Statistics **Exhibit 3**.

## PHASE 4 16:00–18:00 Routine duties (tutoring and Q&A)

9. Provide on-site Q&A and supervise students. The procedure depends on what each student is working on:

### Scenario A. Students studying a Module

1. The student works through the current Module using the recorded lessons, learning materials and worked examples, and completes the corresponding Demo questions.
2. During this study, the tutor answers the student's questions about understanding the concepts and applying them to problems, and observes whether the student can use the methods independently.
3. When the student believes they have mastered the Module, they must complete the designated Demo independently, without looking at the recordings, answers or lecture notes.
4. After the Demo, the tutor checks the student's working, accuracy, choice of method and grasp of the core concepts.
5. If the student meets the standard, schedule them for the corresponding Module Test. If not, point out the specific problems and require the student to re-watch the relevant lesson content, complete corrections or extra practice, and then be reviewed again.

### Scenario B. Students preparing for a SAC

1. First confirm the date of the student's SAC, the topics it covers and any materials provided by the school.
2. Check, against the SAC scope, whether the student has completed the corresponding Modules. If the underlying content has not yet been mastered, prioritise covering the relevant Module content.
3. Once the underlying content is complete, have the student complete a Practice SAC provided by the school or the centre to simulate the question types and requirements of the real assessment.
4. Mark the Practice SAC and analyse the student's main problems based on where marks were lost.
5. Provide on-site Q&A for the concepts or question types that need further work. If there are many issues or a focused explanation is needed, schedule a separate Q&A session.

### Scenario C. Students sitting an exam (Module Tests and VCAA mock exams)

1. The student sits the mock exam.
2. After marking, review the exam with the student and analyse the problems it revealed, including but not limited to:
   - gaps in knowledge;
   - unfamiliarity with solution methods;
   - poor time allocation;
   - carelessness or calculation errors;
   - failure to check answers;
   - answering-strategy problems under formal exam conditions.
3. Provide Q&A for the mistakes the student cannot resolve on their own, and require the student to complete the necessary corrections.

4. Decide the next step based on the exam result, for example: revise the relevant Module, add targeted practice, redo the Demo, schedule a Retest, or move on to the next mock exam.
5. After each exam, require the student to record their personal exam data, including:
   - exam date;
   - Module name, or the year and name of the VCAA paper;
   - start time;
   - end time of the first attempt;
   - end time of the second attempt;
   - number of checks performed;
   - final score.

   *Keeping a continuous record of scores and time taken helps the student observe changes in their knowledge, answering speed and exam performance over time.*

## PHASE 5 18:20–18:30 Daily work report

Have the manager sign the work sheet and give a brief report on the day, covering:

- number of students who studied on site today;
- number of students helped in Q&A, and the approximate number of questions;
- number of test papers marked;
- SAC and Practice SAC answers completed;
- number of parent feedback messages completed;
- students and matters that need priority follow-up.

# 05 EXHIBITS

Samples of the three record sheets. Highlighted tags in Sections 1 and 4 mark where each sheet is used.

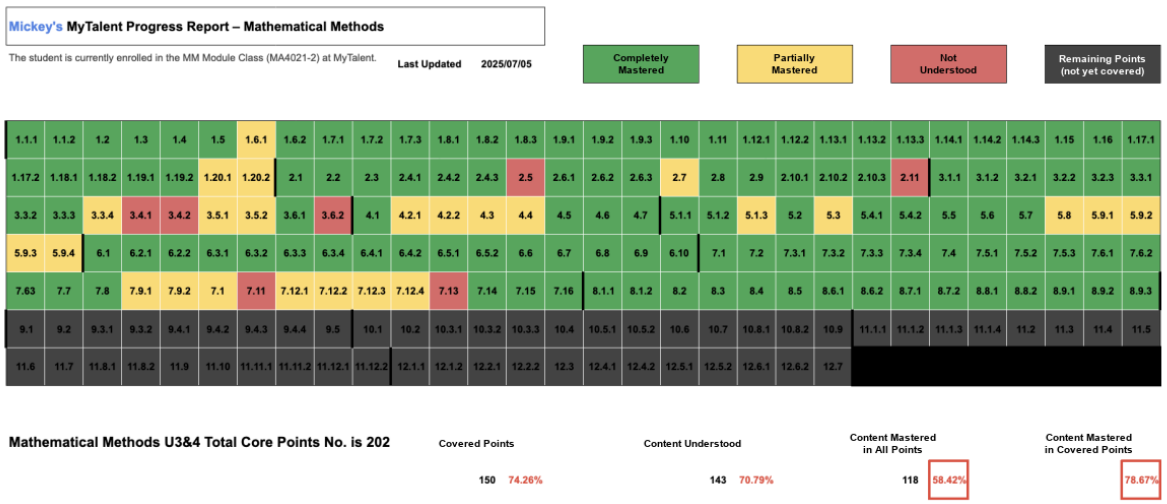
## EXHIBIT 1 Marking Sheet

*Per-test record of each question, the syllabus point it tests, and the outcome (DONE = correct, W = wrong, C = careless error, Blank = not attempted).* ***Used in step 8***

| Module Test 8A | | |
|---|---|---|
| Question | Module 8 | |
| 1 | 8.1 | DONE |
| 2 | 8.2 | W |
| 3 | 8.4 | DONE |
| 4 | 8.5 | DONE |
| 5 | 8.6.2 | DONE |
| 6a | 8.7.1 | DONE |
| 6b | 8.7.2 | DONE |
| 7 | 8.8.1 | C |
| 8 | 8.8.2 | \Blank\ |
| 9 | 8.8.1 | DONE |
| 10 | 8.9.2 | DONE |
| 11 | 8.9.3 | DONE |

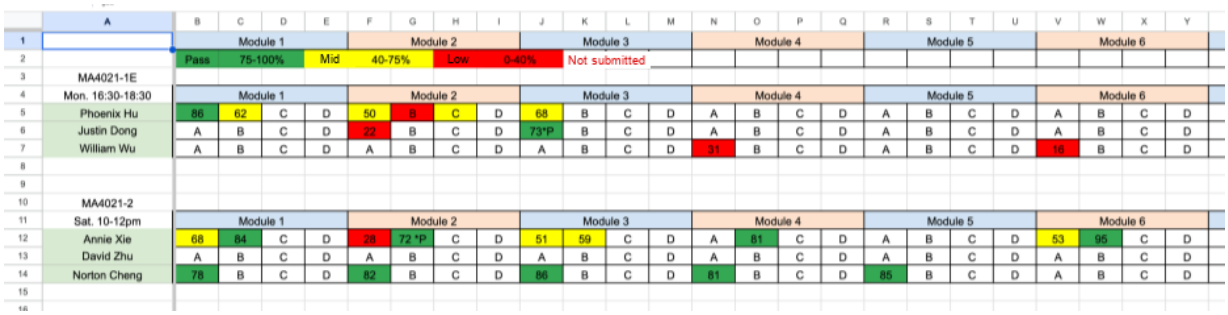
## EXHIBIT 2 Student Progress Sheet

*Heat-map of every core knowledge point in the course, coloured by mastery level (completely mastered, partially mastered, not understood, not yet covered), with summary percentages.* ***Used in steps 1, 4 and 8***




## EXHIBIT 3 Student Exam Statistics

*Class-level spreadsheet of Module Test scores and attempt outcomes for each student, colour-coded by performance band.* ***Used in steps 1 and 8***



| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | Module 1 | | | | Module 2 | | | | Module 3 | | | | Module 4 | | | | Module 5 | | | | Module 6 | | | |
| 2 | | Pass | 75-100% | | Mid | 40-75% | | Low | 0-40% | Not submitted | | | | | | | | | | | | | | | |
| 3 | MA4021-1E | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | Mon. 16:30-18:30 | Module 1 | | | | Module 2 | | | | Module 3 | | | | Module 4 | | | | Module 5 | | | | Module 6 | | | |
| 5 | Phoenix Hu | 86 | 62 | C | D | 50 | B | C | D | 68 | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D |
| 6 | Justin Dong | A | B | C | D | 22 | B | C | D | 73*P | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D |
| 7 | William Wu | A | B | C | D | A | B | C | D | A | B | C | D | 31 | B | C | D | A | B | C | D | 16 | B | C | D |
| 8 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 9 | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10 | MA4021-2 | | | | | | | | | | | | | | | | | | | | | | | | |
| 11 | Sat. 10-12pm | Module 1 | | | | Module 2 | | | | Module 3 | | | | Module 4 | | | | Module 5 | | | | Module 6 | | | |
| 12 | Annie Xie | 68 | 84 | C | D | 28 | 72 *P | C | D | 51 | 59 | C | D | A | 81 | C | D | A | B | C | D | 53 | 95 | C | D |
| 13 | David Zhu | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D |
| 14 | Norton Cheng | 78 | B | C | D | 82 | B | C | D | 86 | B | C | D | 81 | B | C | D | 85 | B | C | D | A | B | C | D |
| 15 | | | | | | | | | | | | | | | | | | | | | | | | | |